

Julien de Saint Angel

AI Project Lead | Data Scientist | PhD in Applied Computer Science

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PROFILE

PhD in applied computer science (2025) with expertise in complex data analysis and predictive modelling. Specialised in Deep Learning, computer vision and agentic AI, I translate business problems into robust, production-ready technical solutions. I draw on my skills in statistical analysis and anomaly detection to turn raw data into strategic decisions.

SKILLS

- **AI / Machine Learning:** CNN, Transformers, autoencoders, hyperspherical models, SVDD, anomaly detection, NLP, LLM, agentic RAG, text-to-SQL.
- **Frameworks & Libraries:** Python, PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, FastAPI, LangGraph.
- **MLOps & Tools:** Git, Docker, Jupyter, Linux, SQL / SQLite, Ollama, MCP, ML pipelines.
- **Mathematics & 3D:** Optimisation, PDEs, conformal geometry, signal processing, Fusion 360, SolidWorks.

PROFESSIONAL EXPERIENCE

AI Project Lead | *Unapei 17, La Rochelle (fixed-term contract)* **Apr. 2026 – present**

Leading and developing an agentic AI platform for HR and accounting data.

- **Autonomous AI agent (natural language → database):** answers HR questions in French by querying the databases itself, with no technical skills required (Reasoning → Action → Observation, schema linking).
- **Privacy & GDPR compliance:** 100% local architecture (Ollama, SQLite, LangGraph) – no sensitive data leaves the organisation; GDPR is built into the core of the design.
- **Multi-database semantic matching:** designed and developed an algorithm that automatically links heterogeneous, unstructured databases (HR, accounting, activity) into a unified, reliable foundation. Auditable architecture: the code computes the facts, the LLM reasons over them – no hallucination, reproducible and traceable results.
- **Visualisation & decision support:** real-time, GDPR-compliant dashboards to guide HR decision-making.

PhD Researcher / R&D Engineer – AI and Machine Learning | *La Rochelle Université (MIA)* **2020 – 2025**

- Created an **initialisation method that makes hyperspherical layers trainable**.
- Proposed an **anomaly detection method derived from SVDD**, robust on real-world data.

Research & Engineering Internships **2015 – 2019**

- **XLIM (UMR 7252):** Characterisation of sports movements using high-speed cameras (key-point tracking).
- **LIENSS (UMR 7266):** Visual tide gauge (reading a tide staff by computer vision).
- **SYRTE (UMR 8630) - Paris Observatory:** Satellite orbit interpolation (numerical simulation of ordinary differential equations).

Mathematics Teacher | *Saint-Exupéry High Schools, La Rochelle* **2015 – 2017**

- Scientific communication, outreach and team management.

R&D & INNOVATION PROJECTS

Nova – Intelligent AI Assistant Agent rentime.orbitalis.fr

- **Calendar & Email Management:** Automatic appointment scheduling and email drafting through interactive two-way voice synthesis.
- **Automated Journal & Monitoring:** Automated tracking of AI-related news and real-time alerts on new developments.

MIA & JobScope – AI Web Platforms orbitalis.fr

- **Conversational agent (LLM + RAG):** Embeddings and vector database for semantic search over CVs and profiles.
- **JobScope:** Automated intelligent semantic matching engine between CVs and job offers.

EDUCATION

PhD in Applied Computer Science | La Rochelle University **2025**

Thesis: Neural networks with hyperspherical layers for anomaly detection.

Master's in Mathematics and Applications | With honours (AB), La Rochelle University **2019**

Maîtrise (M1) in Astronomy & Physics | Paris-Meudon Observatory **2015**

Master's in Mathematics & CAPES (teaching certification) | La Rochelle University **2014**

PUBLICATIONS & MISCELLANEOUS

- **Publications:** *Multi-Spheres Anomaly Detection* (ICMLA 2024), *Improving Multi-Sphere Anomaly Detection* (2025), *Hyperspherical Layers via CGA* (ORASIS 2021).
- **Languages:** French (native), English (B2 written), Spanish & Romanian (B1).
- **Interests:** Astrophotography, bird photography, 3D modelling, video editing (After Effects).